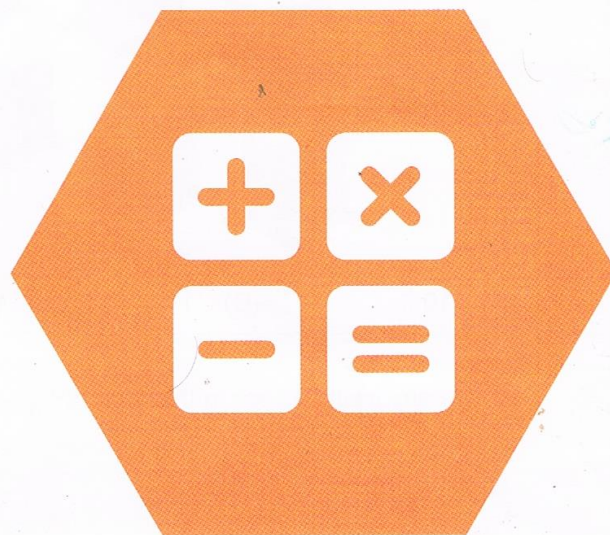




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

INTRODUCTORY PAPER



2014 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 35 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

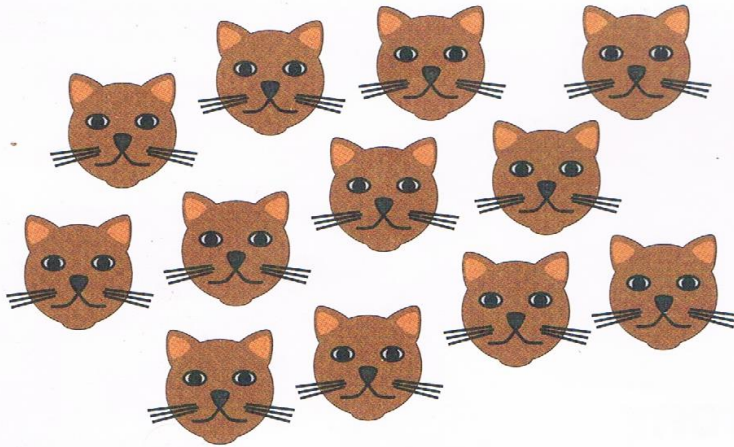
You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

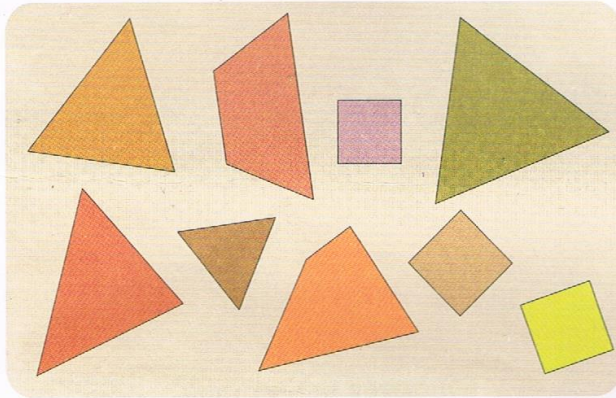
You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

1. How many cats are in this picture?



- (A) 10 (B) 11 (C) 12 (D) 13

2. How many triangles are in this picture?



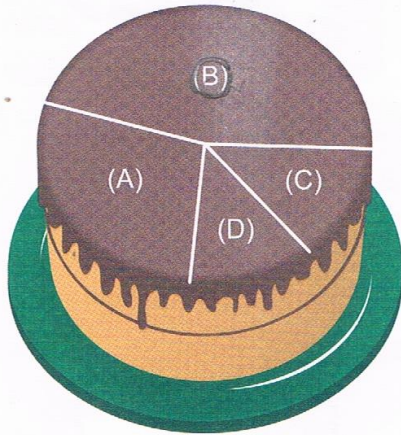
- (A) 4 (B) 5 (C) 7 (D) 8

3. If yesterday was Friday, what day is it tomorrow?

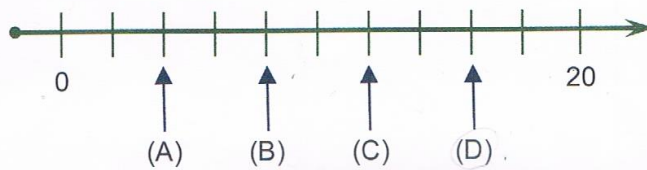
- (A) Thursday (B) Saturday (C) Sunday (D) Monday

4. Amy cut a cake into four unequal pieces.

Which piece was the largest?



5. Which arrow is pointing at 8 on this number line?



6. Sam drew a picture of a bee.



Which of these is also a picture of Sam's bee?



(A)



(B)



(C)



(D)

7. This pattern goes up by 6 each time.

8, 14, 20, 26, 32, ?

What is the next number in this pattern?

- (A) 33 (B) 34 (C) 36 (D) 38

8. This chart shows the shells some children collected at the beach.

Name	Shells
Peter	
Heidi	
Sue	
Henry	

How many shells did Sue collect altogether?

- (A) 4 (B) 5 (C) 6 (D) 8

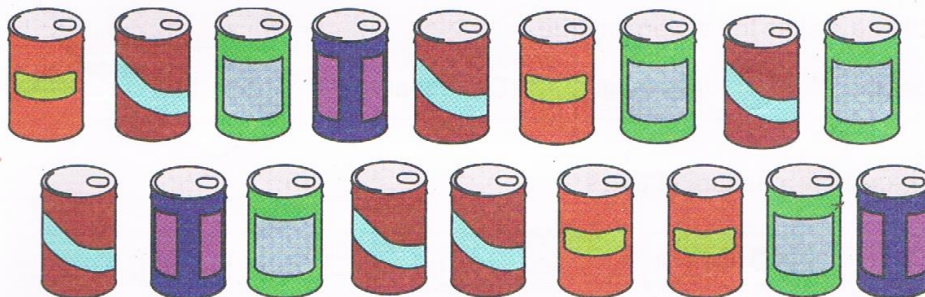
- 9.

 = 5  = 7

 +  +  +  +  = ?

- (A) 12 (B) 14 (C) 15 (D) 29

10. Ella picks one of these cans without looking.



Which kind of can is she most likely to pick?



(A)



(B)

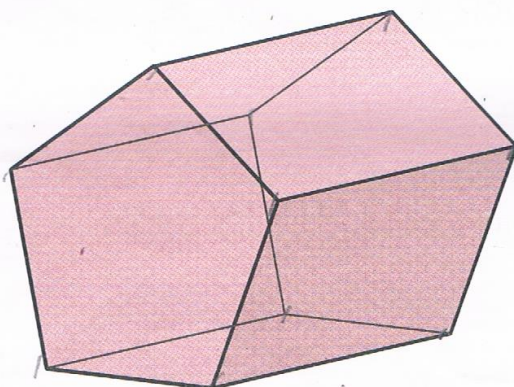


(C)



(D)

11. How many edges does this object have?



(A) 5

(B) 7

(C) 10

(D) 15

12. Jim drew some pictures.

Which of these is **impossible** for him to draw?

- (A) a cat with three legs
- (B) a square with five sides
- (C) a table with five legs
- (D) a car with three wheels

13. $24 - 9$ is 15 17 $8 + 9$

Which of these makes the statement true?

- (A) bigger than (B) less than (C) equal to (D) half of

14. These clocks show a pattern of times.

10:30^{AM}

12:30^{PM}

?

04:30^{PM}

06:30^{PM}

Which of these clocks must replace the ? to fit the pattern?

11:30^{AM}

01:30^{PM}

02:30^{PM}

03:30^{PM}

(A)

(B)

(C)

(D)

15. Toby has 21 apples. Lillian has 4 apples less than Toby.

How many apples does Lillian have?

(A) 4

(B) 17

(C) 18

(D) 25

16. Sue is 180 cm tall. Mel is half as tall as Sue.

How tall is Mel?

(A) 40 cm

(B) 90 cm

(C) 100 cm

(D) 140 cm

17. $120 - 11 = ?$

(A) 131

(B) 119

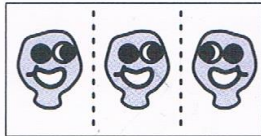
(C) 111

(D) 109

18. Sarah used this face to make a pattern. She flipped it over the dotted lines.



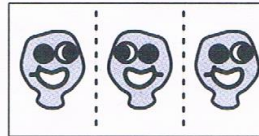
Which picture shows the result?



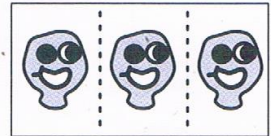
(A)



(B)



(C)



(D)

19. The table shows some information about the books some children read.

	Non-fiction	Fiction
Boys	20	16
Girls	5	15

How many children read fiction books?

- (A) 15 (B) 16 (C) 31 (D) 56

20. Some students were walking along a track.

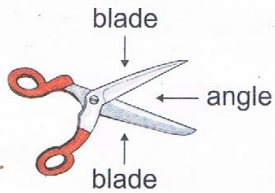
The table shows how far they had walked after 10 minutes.

Student	Distance
Sally	605 metres
Joe	575 metres
Nadia	520 metres
Billy	580 metres
Ellen	610 metres
Alex	565 metres

Who was halfway between Nadia and Ellen?

- (A) Sally (B) Joe (C) Billy (D) Alex

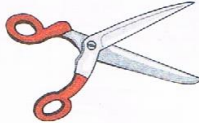
21. The blades of scissors can make an angle.



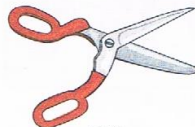
Which of these has the largest angle between its blades?



(A)



(B)

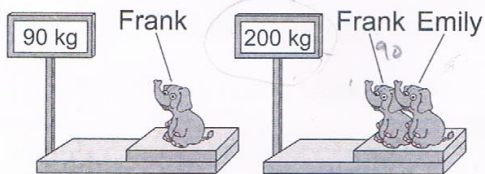


(C)



(D)

22. Emily and Frank are two baby elephants.

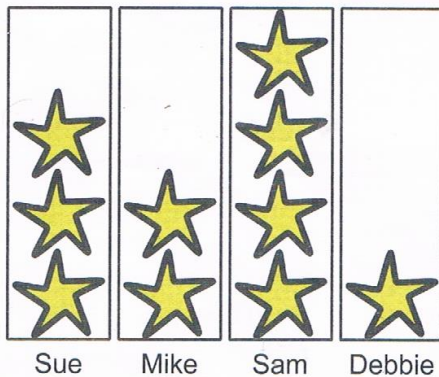


What is Emily's mass?

- (A) 90 kilograms (B) 110 kilograms (C) 190 kilograms (D) 200 kilograms

23. In Mr Zu's class you get a silver star for doing good work. If you have five silver stars, you get a gold star.

Gold Stars



Who has 15 silver stars?

- (A) Sue (B) Mike (C) Sam (D) Debbie

24.

$$\boxed{? + 15 = 28 - 8}$$

What is the missing number?

- (A) 5 (B) 10 (C) 20 (D) 35

25. Each week Anna saves \$10 of her pocket money.

Each week Tim gets \$16 pocket money and saves half of it.

In eight weeks, how much money do Anna and Tim save altogether?

- (A) \$208 (B) \$144 (C) \$88 (D) \$18

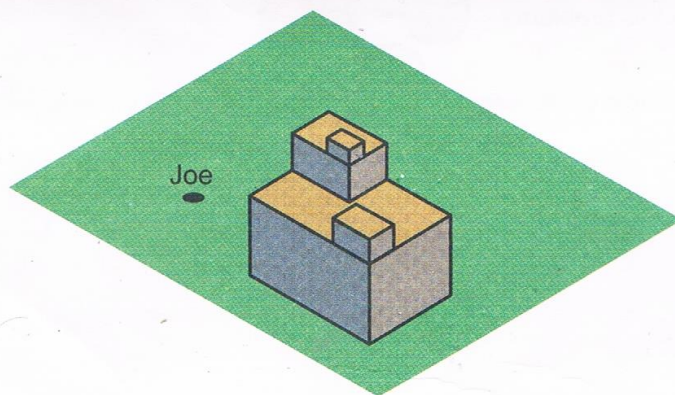
26. Abi is 1 year and 9 months older than Ben.

Ben is 2 years and 8 months younger than Cam.

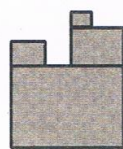
What is the difference in age between Abi and Cam?

- (A) 11 months (B) 13 months
(C) 4 years and 5 months (D) 4 years and 7 months

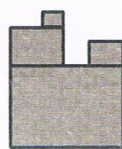
27. Joe is standing near a building as shown.



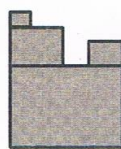
Which of these shows Joe's view of the building?



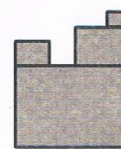
(A)



(B)



(C)

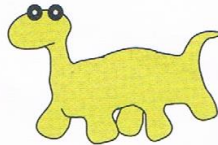


(D)

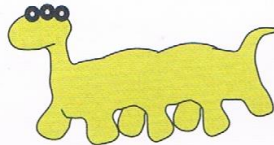
28. Jane is playing a video game. In the game there are monsters of different sizes.



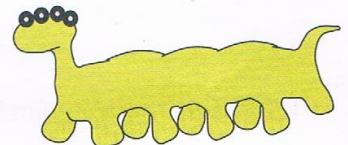
1 eye
3 legs



2 eyes
5 legs



3 eyes
7 legs



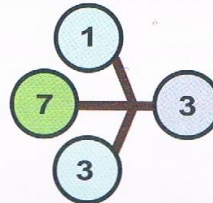
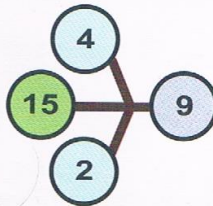
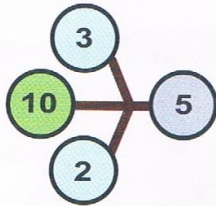
4 eyes
9 legs

Bigger monsters follow the same pattern.

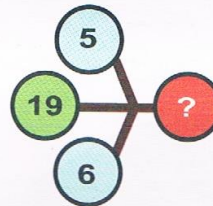
How many eyes would a monster with 25 legs have?

- (A) 12 (B) 13 (C) 14 (D) 15

29. Cindy has a special way of writing number problems.



What is the missing number in this problem?



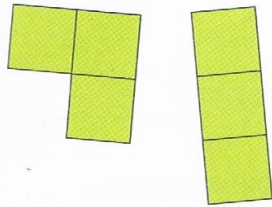
- (A) 8 (B) 11 (C) 13 (D) 14

30. Finn has some square tiles.

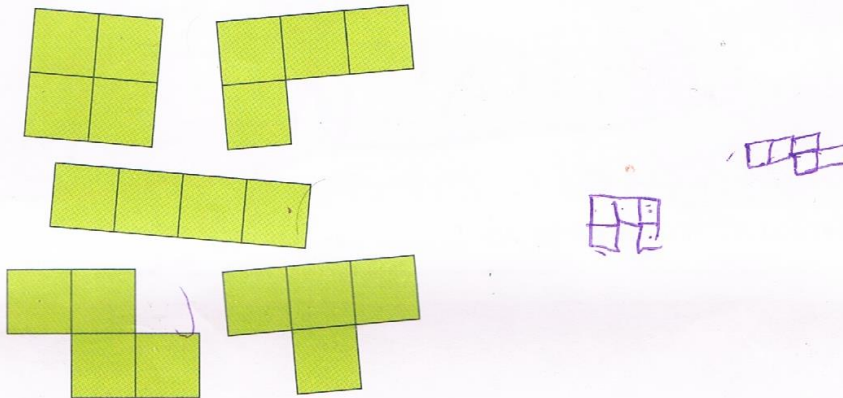
He uses two tiles together. He can only make this shape.



He uses three tiles together. He can only make these two different shapes.



He uses four tiles together. He can only make these five different shapes.



How many different shapes can Finn make using 5 tiles?

- (A) 6
- (B) 10
- (C) 12
- (D) 16